Amit Nath Gupta

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Eligible to work in the UK under the Graduate Route from June 2028

Education

MSc Robotics with Extended Research — **Sep 2026 – Jun 2028**
The University of Manchester — *Manchester, UK*

B.Tech in Electronics and Communication Engineering — **2018 – 2022**
Birla Institute of Technology, Mesra — *CGPA: 8.54/10*

Technical Skills

Languages: Python, C++, Java, JavaScript, Bash, MATLAB
Robotics: ROS/ROS 2, Navigation2, SLAM (Cartographer, Gmapping), Motion Planning, Behavior Trees, Sensor Fusion
Embedded & Systems: Embedded Linux (Ubuntu/Tizen), Firmware, UART/Serial Communication, HDMI-CEC, KASAN, AddressSanitizer (ASan), Memory Profiling, Crash Analysis, RPM, Perforce Helix Core (P4), Git, Docker
Testing & Release: Release Management, Software Integration, Test Automation, Regression, Sanity & Long-run Testing, Field & Compatibility Testing, Defect Triage, Root Cause Analysis, Jira
AI & Computer Vision: OpenCV, Scikit-Learn, TensorFlow, Keras, EasyOCR, Signal Processing
Tools & Libraries: RViz, Gazebo, Splunk, Tableau, PyAutoGUI, pyserial, API Integration

Projects

Simultaneous Localization and Mapping (SLAM) using TortoiseBot — **May 2025 – Sep 2025**
Executive Programme in Robotics, IIT Delhi

- Completed full hardware assembly integrating an **8 GB Raspberry Pi, YDLiDAR S2, IMU, motor driver and Raspberry Pi camera**, then deployed **teleoperation and autonomous navigation** using ROS packages.
- Integrated and interfaced **LiDAR and camera sensors** with actuators, applying **sensor fusion** techniques.
- Resolved build conflicts across the **Navigation2, Cartographer and BehaviorTree** ROS packages.
- Diagnosed and fixed a critical **RViz** defect where LiDAR points persisted in a circular pattern, duplicating obstacles by generating redundant data points along the robot's traversed path.

Experience

Samsung R&D Institute India-Delhi (SRI-Delhi) — **July 2022 – Aug 2026**
Software Engineer — *Noida, India*

- During early commercialization on unstable images, ran kernel & user space sanitizers — **KASAN** and **AddressSanitizer (ASan)** — flashed on target hardware and EVK evaluation boards to identify **heap-use-after-free**, buffer overflows, memory leaks, bugs and crashes on widely used applications such as Netflix and on drivers in advance, monitoring kernel logs for tainted states; correlated each **physical action performed on the hardware** — HDMI attach-detach, volume control, app launches — with the **driver and kernel-level fault it triggered**.
- Optimized Smart TV platforms for low memory configurations under **constrained RAM environments**; system-level debugging with **memory optimization and footprint reduction**, performance tuning and feature stability. Worked on a daemon that checks CPU, flash and RAM usages of all processes running inside the TV product and cleans them on excessive usage.
- **Memory profiling** of every binary before release in market to optimize the TV performance for the end user; debugged and resolved memory-related **crashes, leaks, deadlocks** surfaced in long-run testing and performance bottlenecks through crash dump and call stack analysis.
- **Release and hotfix management** of critical Tizen OS defects across commercialization and post-market OTA update cycles: used **P4 (Perforce Helix)** to build full and hotfix TV firmware images, manage code submissions, compare code and identify package dependencies; triaged defects by severity and component, verified each fix on a private image before acceptance, and cherry-picked **changelists** into release images, owning software integration across **5 chipsets** for ATSC, DVB and ISDB regions and **350+ fixes** with **95%** within target schedule.
- Analysed **serial and HyperUART logs** from performance images to narrow down problems based on analysis of logs and code; debugged **operating time performance** issues — boot time, power efficiency — and **user experience performance** issues including UI lag and responsiveness delays, measured per process in milliseconds against specification.
- Built a **Java** application integrating APIs supplied by Samsung HQ to auto-create linked child defect records in the internal defect tracking system, generating 40–200 records in seconds where the stock application crashed beyond 10–15; built **Python** test automation tools (pyserial, regex, multithreading) for pre-production checks over serial, changelist verification by parsing firmware XML, and data validation on hotfix images.

- Delivered the complete **product validation and product release** for a premium North American Smart TV model (**Black Friday launch**) **single-handedly**, a scope usually staffed by three to four engineers: drove the QA meetings and regression rounds, planned **test-schedule risk mitigation** across India and Korea holiday periods, and escalated open UEP/OTP defects to Samsung HQ for spec decisions. On the final factory image trigger, caught an **outdated configuration data file** from its filename, coordinated the fix and the root cause analysis across HQ and developers within hours, and delivered final production firmware to the factory within the deadline.
- Designed and built a **firmware image registration automation tool** for factory mass production, **adopted across the TV Product Group**: an OCR, OpenCV and PyAutoGUI uploader paired with an **independent reviewer that verifies every image by cryptographic checksum**, running 48–60 image sets per cycle **unattended overnight**, reducing a four-person, two-hour process to a single operator, and detecting upload errors across the team before they reached mass production of 5,000–10,000-unit batches. **Led a team of three interns** extending it — assigned and tracked work through Jira cards, reviewed their code, and guided the React front-end build on top of my Python codebase — and trained a fourth intern from scratch on log analysis and defect triage for a commercialization project.
- Acted as the primary liaison between the SRI-Delhi Product group and QA, and between SRI-Delhi and Samsung HQ Korea, to successfully execute pre-production shipments: walked testers through HQ change requests and their impact, coordinated field testers across five regions, negotiated release scope and deferrals, and managed schedule dependencies on third-party app releases.

Samsung R&D Institute India-Delhi (SRI-Delhi) **January 2022 – July 2022**
Intern *Noida, India*

- Built a web tool (**HTML, CSS, JavaScript, Python CGI**) for developers to unlock password-protected TV log files: the user enters the passcode for a group of logs and uploads any number of files, and the script unlocks them into a separate directory.
- Built a **JavaScript web application** for preparing spec sheets: users upload any number of screenshots dynamically, select an image and tag it from a drop-down menu, and save every image with its tag to a database for year-on-year spec comparison.
- Verified developer fixes with Linux **RPM** packages — flashing an RPM built from a shelved changelist onto a debug image through shell commands and following the defect's reproduction route, confirming the fix without a full image rebuild.

Publications & Conference Papers

Author name shown in bold.

- Kaur, P. & **Gupta, A. N.** (2025). A Medical Diagnosis of Cataract Using Intuitionistic Fuzzy Sets. *Risk Assessment and Management Decisions*, 2(2), 88–94. **DOI:** 10.48314/ramd.vi.63
- Kaur, P. & **Gupta, A. N.** (2025). Circular Economy Business Model Selection Using MCDM Methods (MOORA and EDAS). *Proceedings of the 1st International Conference on Future Business Perspectives*, Nguyen Tat Thanh University, pp. 377–387. **ISBN:** 978-1-9193829-0-6
- Kaur, P. & **Gupta, A. N.** (2025). Applying Hybrid ML-MCDM Approaches for Ranking Jharkhand Districts for Sustainable Mining. *8th International Conference on Entrepreneurship for Sustainability & Impact*, Qatar University, Doha.
- Kaur, P. & **Gupta, A. N.** (2019). A Comparative Analysis of a Multi-Objective Investment Selection Problem Using Fuzzy Logic. *International Conference on Operations Research and Decision Sciences*, IIM Visakhapatnam.